MESH-ONLY CONVERGENCE AND SHARPER RATES ON ORTHOGONAL TILINGS

Abstract. We prove mesh-only convergence for the discrete Dirichlet problem on orthogonal tilings. If the maximum tile diameter is ϵ, then smooth harmonic data have uniform extension error at most $C\|h\|_{C^2}\epsilon^{4/(d+4)}$, giving exponent $2/3$ in dimension two. The proof uses tile-volume reversible time, a defective Nash inequality, heat-kernel smoothing, and a local power potential. Spectral contraction bounds the expected exit time of a small-volume set; coordinate martingales and the exact clock identity then turn the weighted L^2 error into the uniform bound. Under the remaining domain hypotheses of Bou-Rabee and Gwynne, the same mesh-only argument gives convergence of harmonic measures and stopped canonical walks modulo time parametrization.

Draft proof note

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1. Statement of the result

We use the notation of Bou-Rabee and Gwynne [2, Sections 1.1 and 2.1]. Let $D \subset \mathbb{R}^d$ be open. An *orthogonal tiling* of D is a locally finite collection $(P_v)_{v\in\mathcal{V}}$ of closed convex polytopes with pairwise disjoint interiors, together with marked points $v \in \operatorname{int} P_v$, such that the polytopes cover D. Distinct vertices $v, w \in \mathcal{V}$ are joined by an edge when

$$H_e := P_v \cap P_w, \qquad e = \{v, w\},$$

is a $(d-1)$-dimensional facet. We assume that $w - v$ is orthogonal to the affine span of H_e. Write

$$m_v := \operatorname{vol}_d(P_v), \qquad b(e) := \operatorname{vol}_{d-1}(H_e), \qquad \|e\| := \|w-v\|, \qquad a(e) := \frac{b(e)}{\|e\|}. \tag{1}$$

Date: October 11, 2026.

The weighted adjacency graph is denoted by $\mathcal{G} = (\mathcal{V}, \mathcal{E})$. Its mesh is

$$\epsilon := \sup_{v \in \mathcal{V}} \operatorname{diam}(P_v). \tag{2}$$

For $U \Subset D$, let

$$\mathcal{V}[U] := \mathcal{V} \cap U, \qquad \partial\mathcal{V}[U] := \{w \in \mathcal{V} \setminus \mathcal{V}[U] : w \sim v \text{ for some } v \in \mathcal{V}[U]\},$$

and set

$$\overline{\mathcal{V}[U]} := \mathcal{V}[U] \cup \partial\mathcal{V}[U].$$

We write $\overline{\mathcal{E}[U]}$ for the set of edges with at least one endpoint in $\mathcal{V}[U]$, and

$$\widehat{U} := \bigcup_{v \in \overline{\mathcal{V}[U]}} P_v. \tag{3}$$

Local finiteness implies that $\overline{\mathcal{V}[U]}$ is finite whenever U is bounded and $\overline{U} \subset D$.

For a function $f : \mathcal{V} \to \mathbb{R}$, the weighted discrete gradient and Laplacian are

$$\nabla_a^{\mathcal{G}} f(v, w) := a(v, w)(f(w) - f(v)), \tag{4}$$

$$\Delta_a^{\mathcal{G}} f(v) := \sum_{w \sim v} a(v, w)(f(w) - f(v)). \tag{5}$$

If f vanishes outside a finite set, its Dirichlet energy and its weighted L^p norms are

$$\mathcal{D}(f) := \sum_{e=\{v,w\} \in \mathcal{E}} a(e)(f(w) - f(v))^2, \qquad \|f\|_{p,m}^p := \sum_{v \in \mathcal{V}} m_v |f(v)|^p. \tag{6}$$

For a function on $\overline{\mathcal{V}[U]}$ which vanishes on $\partial\mathcal{V}[U]$, we extend it by zero to $\mathcal{V} \setminus \overline{\mathcal{V}[U]}$. With this convention, $\mathcal{D}(f)$ agrees with the energy $\mathcal{D}(f, U)$ of Bou-Rabee and Gwynne [2, Equation (2.7)].

Let h_C be harmonic and C^2 on an open neighborhood of $\overline{U}$. Whenever every connected component of $\mathcal{V}[U]$ is joined to $\partial\mathcal{V}[U]$, its discrete Dirichlet extension is the function $h_D : \overline{\mathcal{V}[U]} \to \mathbb{R}$ satisfying

$$\begin{cases} \Delta_a^{\mathcal{G}} h_D = 0 & \text{on } \mathcal{V}[U], \\ h_D = h_C & \text{on } \partial\mathcal{V}[U]. \end{cases} \tag{7}$$

The following finite-network fact will be used repeatedly.

Lemma (Dirichlet principle and maximum principle). Let A be a finite set of vertices, let ∂A be a finite set of neighboring boundary vertices, and suppose that every connected component of A is joined by an edge path in $A \cup \partial A$ to ∂A. For prescribed boundary values $g : \partial A \to \mathbb{R}$, there is a unique function $u : A \cup \partial A \to \mathbb{R}$ which equals g on ∂A and satisfies $\Delta_a^{\mathcal{G}} u = 0$ on A. Moreover,

$$\min_{\partial A} g \le u(v) \le \max_{\partial A} g, \qquad v \in A \cup \partial A. \tag{8}$$

Proof. On the affine space of functions equal to g on ∂A, minimize

$$\frac{1}{2} \sum_{\{v,w\} :\, \{v,w\} \cap A \neq \varnothing} a(v, w)(u(v) - u(w))^2.$$

A minimizer exists because the space is finite dimensional. The path-to-boundary assumption makes the quadratic part positive definite on functions vanishing on ∂A, so the minimizer is unique. Its Euler–Lagrange equations are $\Delta_a^{\mathcal{G}} u = 0$ on A.

For the maximum principle, suppose that u attains its maximum at an interior vertex v. Harmonicity gives

$$u(v) = \frac{1}{a(v)} \sum_{w \sim v} a(v,w)u(w), \qquad a(v) := \sum_{w \sim v} a(v,w).$$

Since every weight is positive, every neighbor of v must have the same value. Repeating this argument along an edge path to the boundary shows that the maximum is attained on ∂A. Applying the same argument to $-u$ gives the minimum bound. □

We next verify the path-to-boundary condition in the small-mesh regime needed below. Since $\overline{U} \Subset D$, choose $\delta > 0$ such that $B_\delta(\overline{U}) \subset D$. The number of connected components of D which meet $\overline{U}$ is finite: if one chooses a point of $\overline{U}$ in each such component, the corresponding balls of radius $\delta/2$ are pairwise disjoint and lie in the bounded set $B_\delta(\overline{U})$. Denote these components by $D_1, \dots, D_N$. For each j, the compact set $\overline{U} \cap D_j$ is a proper subset of D_j, so we may choose

$$y_j \in D_j \setminus \overline{U}, \qquad \rho_j := \operatorname{dist}(y_j, \overline{U}) > 0.$$

Set $\rho_U := \min_{1 \le j \le N} \rho_j$. Suppose that $\epsilon < \rho_U$, and fix $v \in \mathcal{V}[U]$. Let D_j be the component containing v, and choose a cell P_{w_j} containing y_j. Since $\|w_j - y_j\| \le \operatorname{diam}(P_{w_j}) \le \epsilon < \rho_j$, one has $w_j \notin U$.

A polygonal path in D_j joins v to w_j. Its image is compact and contained in the open set D_j, so it has a compact tubular neighborhood contained in D_j. By local finiteness, only finitely many cells meet this neighborhood. Perturbing the path inside the neighborhood, we may arrange that it avoids every face of dimension at most $d-2$ of these finitely many cells and crosses each encountered facet transversely. Reading the cells in the order in which the perturbed path visits them gives an edge path in the facet-adjacency graph from v to w_j. The first marked point on this edge path which does not belong to U is a vertex of $\partial\mathcal{V}[U]$. Thus every connected component of $\mathcal{V}[U]$ is joined to $\partial\mathcal{V}[U]$ whenever $\epsilon < \rho_U$, and the finite-network lemma above gives existence and uniqueness for (7) in this regime.

Theorem 1.1 (Mesh-only Dirichlet convergence)**.** *Let $d \ge 2$, let $U \Subset D$ be bounded, and let $(\mathcal{G}_n)_{n \ge 1}$ be a sequence of orthogonal tilings of D whose mesh sizes ϵ_n tend to zero. Let h_C be C^2 and harmonic on a neighborhood of $\overline{U}$, and, for all sufficiently large n, let $h_{D,n}$ solve (7) on $\mathcal{G}_n$. Then*

$$\lim_{n \to \infty} \max_{v \in \mathcal{V}_n[U]} |h_{D,n}(v) - h_C(v)| = 0. \tag{9}$$

More precisely, if K is a fixed bounded open set such that $\overline{U} \subset K \Subset D$ and h_C is C^2 and harmonic on a neighborhood of $\overline{K}$, then, for all sufficiently small ϵ_n,

$$\max_{v \in \mathcal{V}_n[U]} |h_{D,n}(v) - h_C(v)| \le C\epsilon_n^{\beta_d}, \qquad \beta_d := \frac{4}{d+4}, \tag{10}$$

where C depends only on d, U, K, and $\|h_C\|_{C^2(K)}$. In particular, the convergence is uniform over families for which these quantities are uniformly bounded.

The process introduced in section 3 is only an auxiliary clock. Its embedded jump chain has transition probabilities

$$\mathbb{P}[v \to w] = \frac{a(v,w)}{\sum_{u \sim v} a(v,u)},$$

which are exactly those of the canonical discrete-time walk in Bou-Rabee and Gwynne [2, Equation (1.1)]. Changing the holding times therefore does not change harmonic functions, exit locations, or harmonic measure.

Corollary 1.2. *In the orthogonal-tiling setting of Bou-Rabee and Gwynne [2, Theorems A and B], the mesh hypothesis alone verifies Assumption D of Bou-Rabee and Gwynne [2, Section 4]. Consequently, the alternatives (I), (II), and (III) used there to establish Assumption D are unnecessary: the harmonic-measure conclusion of Bou-Rabee and Gwynne [2, Proposition 4.2] and the stopped-walk convergence conclusion of Bou-Rabee and Gwynne [2, Theorem 4.1] hold under the remaining hypotheses.*

2. A defective Nash inequality

Throughout this section, $\mathcal{G}$ is a fixed orthogonal tiling with mesh $\epsilon < \infty$. Every function under consideration has finite support and satisfies

$$\bigcup_{v:\, u(v)\neq 0} P_v \Subset D. \tag{11}$$

This is the only localization needed in the analytic estimates. In the application, the support is contained in $\mathcal{V}[U]$ and (11) follows from $U \Subset D$ once the mesh is sufficiently small. Under (11), the piecewise-constant extension used below vanishes in a neighborhood of ∂D, so its zero extension to $\mathbb{R}^d$ creates no ambient-boundary variation.

Fix an edge $e = \{v, w\}$. Let $h_{v,e}$ and $h_{w,e}$ be the Euclidean distances from v and w, respectively, to the hyperplane containing H_e. The marked points lie on opposite sides of this hyperplane and $w - v$ is normal to it, so

$$\|e\| = h_{v,e} + h_{w,e}. \tag{12}$$

The pyramids $\mathrm{Conv}(v, H_e)$, as e ranges over the edges incident to v, partition P_v up to sets of d-dimensional measure zero. Therefore

$$\sum_{e\ni v} b(e) h_{v,e} = d m_v. \tag{13}$$

This is the same pyramid decomposition used in Bou-Rabee and Gwynne [2, Lemma 2.4].

Define an auxiliary vertex measure by

$$\nu_v := \frac{1}{2}\sum_{e\ni v} a(e)\|e\|^2 = \frac{1}{2}\sum_{e\ni v} b(e)\|e\|. \tag{14}$$

Lemma 2.1 (Comparison of ν with tile volume)**.** *For every finitely supported $u : \mathcal{V} \to \mathbb{R}$,*

$$\sum_{v\in\mathcal{V}} \nu_v u(v)^2 \le \frac{3d}{2}\|u\|_{2,m}^2 + 4\epsilon^2 \mathcal{D}(u). \tag{15}$$

Proof. By (12), (13), and (14),

$$\begin{aligned}\sum_v \nu_v u(v)^2 &= \frac{1}{2}\sum_v \sum_{e=\{v,w\}\ni v} b(e)(h_{v,e} + h_{w,e})u(v)^2 \\ &= \frac{d}{2}\sum_v m_v u(v)^2 + \frac{1}{2}\sum_v \sum_{e=\{v,w\}\ni v} b(e) h_{w,e} u(v)^2.\end{aligned}$$

For each oriented incidence (v,e) with $e=\{v,w\}$,

$$u(v)^2 \le 2u(w)^2 + 2(u(v)-u(w))^2.$$

Substitution into the second double sum gives

$$\sum_v \nu_v u(v)^2 \le \frac{d}{2}\|u\|_{2,m}^2 + \sum_v \sum_{e=\{v,w\}\ni v} b(e)h_{w,e}u(w)^2 + \sum_v \sum_{e=\{v,w\}\ni v} b(e)h_{w,e}(u(v)-u(w))^2.$$

In the first double sum on the right, collect terms according to w and use (13); it equals $d\|u\|_{2,m}^2$. In the second double sum, the two orientations of each edge combine, by (12), to give

$$\sum_{e=\{v,w\}} b(e)\|e\|(u(v)-u(w))^2 = \sum_{e=\{v,w\}} a(e)\|e\|^2(u(v)-u(w))^2.$$

If $y\in H_e$, then

$$\|e\| = \|w-v\| \le \|w-y\| + \|y-v\| \le \operatorname{diam}(P_w) + \operatorname{diam}(P_v) \le 2\epsilon.$$

The last display is therefore at most $4\epsilon^2\mathcal{D}(u)$, which proves (15). □

Set

$$q := \frac{2d}{d-1}. \tag{16}$$

Lemma 2.2 (BV–Sobolev inequality)**.** *There is a constant $C_d<\infty$ such that every finitely supported $u:\mathcal{V}\to\mathbb{R}$ satisfies*

$$\|u\|_{q,m}^2 \le C_d\mathcal{D}(u)^{1/2}\big(\|u\|_{2,m}^2 + \epsilon^2\mathcal{D}(u)\big)^{1/2}. \tag{17}$$

Proof. Extend u by zero to all vertices and define the piecewise-constant function

$$F(y) := u(v)^2 \qquad \text{for a.e. } y\in P_v. \tag{18}$$

Since u has finite support, F has compact support in D and its zero extension belongs to $BV(\mathbb{R}^d)$. We first record its total variation. Cellwise integration by parts against a vector field $\varphi\in C_c^1(\mathbb{R}^d;\mathbb{R}^d)$ gives

$$\int_{\mathbb{R}^d} F\,\operatorname{div}\varphi\,\mathrm{d}y = \sum_v u(v)^2\int_{P_v}\operatorname{div}\varphi\,\mathrm{d}y = \sum_{e=\{v,w\}}(u(v)^2-u(w)^2)\int_{H_e}\varphi\cdot n_{v,w}\,\mathrm{d}\mathcal{H}^{d-1},$$

where $n_{v,w}$ is the unit normal to H_e pointing from P_v into P_w. There is no contribution from ∂D because F vanishes in a neighborhood of it. Taking the supremum over $\|\varphi\|_\infty\le 1$ yields

$$|DF|(\mathbb{R}^d) \le \sum_{e=\{v,w\}} b(e)|u(v)^2-u(w)^2|. \tag{19}$$

In fact equality holds, but the upper bound is all that is needed.

The Euclidean BV–Sobolev inequality [1, Theorem 3.47] states that

$$\|F\|_{L^{d/(d-1)}(\mathbb{R}^d)} \le C_d|DF|(\mathbb{R}^d). \tag{20}$$

The left side equals $\|u\|_{q,m}^2$. For the right side, use

$$|u(v)^2 - u(w)^2| = |u(v) - u(w)|\,|u(v) + u(w)|$$

and Cauchy–Schwarz to obtain

$$\sum_e b(e)|u(v)^2 - u(w)^2| \le \left(\sum_e a(e)(u(v) - u(w))^2\right)^{1/2} \times \left(\sum_e b(e)\|e\|(|u(v)| + |u(w)|)^2\right)^{1/2}.$$

Since $(|s| + |t|)^2 \le 2s^2 + 2t^2$ and $\sum_{e\ni v} b(e)\|e\| = 2\nu_v$, the second squared factor is at most

$$4\sum_v \nu_v u(v)^2.$$

Consequently, Lemma 2.1 gives

$$|DF|(\mathbb{R}^d) \le C_d \mathcal{D}(u)^{1/2}\big(\|u\|_{2,m}^2 + \epsilon^2 \mathcal{D}(u)\big)^{1/2}.$$

Combining this estimate with (20) proves (17). □

Proposition 2.3 (Defective Nash and support inequalities)**.** *There is a constant $C_d < \infty$ such that every finitely supported $u : \mathcal{V} \to \mathbb{R}$ satisfies*

$$\|u\|_{2,m}^{4+4/d} \le C_d \|u\|_{1,m}^{4/d} \mathcal{D}(u)\big(\|u\|_{2,m}^2 + \epsilon^2 \mathcal{D}(u)\big). \tag{21}$$

If

$$V_u := \sum_{v : u(v) \neq 0} m_v, \tag{22}$$

then

$$\|u\|_{2,m}^2 \le C_d\big(V_u^{2/d} + \epsilon V_u^{1/d}\big)\mathcal{D}(u). \tag{23}$$

Proof. The log-convexity interpolation inequality for L^p norms, in the form stated in Folland [4, Section 6.5], gives

$$\|u\|_{2,m} \le \|u\|_{1,m}^{1/(d+1)} \|u\|_{q,m}^{d/(d+1)}. \tag{24}$$

Indeed,

$$\frac{1}{2} = \frac{1}{d+1} \cdot 1 + \frac{d}{d+1} \cdot \frac{1}{q}, \qquad q = \frac{2d}{d-1}.$$

Combining (24) with (17), and then raising both sides to the power $2(d+1)/d$, gives (21).

For the support estimate, Hölder's inequality gives

$$\|u\|_{2,m}^2 \le V_u^{1/d} \|u\|_{q,m}^2. \tag{25}$$

Let $X := \|u\|_{2,m}^2$ and $Y := \mathcal{D}(u)$. From (25) and (17),

$$X^2 \le C_d V_u^{2/d} Y(X + \epsilon^2 Y). \tag{26}$$

If $Y = 0$, then (26) gives $X = 0$. Suppose $Y > 0$ and set $R := X/Y$. Dividing (26) by Y^2 gives

$$R^2 \le C_d V_u^{2/d}(R + \epsilon^2).$$

The positive root of the corresponding quadratic is bounded by

$$C_d V_u^{2/d} + C_d \epsilon V_u^{1/d}.$$

Since $X = RY$, this proves (23). □

3. Heat-kernel smoothing in tile-volume time

To avoid any issue concerning explosion of a process on an infinite graph, all processes below are first defined on a finite vertex set. Let $W \subset \mathcal{V}$ be finite and assume that every neighbor of a vertex in W lies in a finite set $\overline{W}$. Declare the vertices of $\overline{W} \setminus W$ absorbing. On W, define the continuous-time generator

$$(\mathcal{L}f)(v) := \frac{1}{m_v}\Delta_a^{\mathcal{G}} f(v) = \frac{1}{m_v}\sum_{w\sim v} a(v,w)(f(w) - f(v)). \tag{27}$$

Thus the jump rate from v to w is $a(v,w)/m_v$. The total holding time at v is exponential with mean

$$\frac{m_v}{a(v)}, \qquad a(v) := \sum_{w\sim v} a(v,w),$$

and, conditional on jumping, the next vertex is w with probability $a(v,w)/a(v)$. The measure m is reversible for the process.

Let $S \subset W$ and let

$$\tau_S := \inf\{t \geq 0 : \widetilde{X}_t \notin S\}.$$

Write $p_t^S(x,y)$ for the transition density of the process killed on exiting S, with respect to m:

$$\mathbb{P}_x[\widetilde{X}_t = y,\ t < \tau_S] = m_y p_t^S(x,y). \tag{28}$$

Since the killed state space is finite, its semigroup is a differentiable self-adjoint contraction semigroup on $L^2(S,m)$. In particular,

$$p_t^S(x,y) = p_t^S(y,x), \qquad p_{s+t}^S(x,y) = \sum_{z\in S} m_z p_s^S(x,z) p_t^S(z,y). \tag{29}$$

These are the standard finite-state continuous-time Markov-chain identities; see, for example, Norris [5, Chapter 2].

Proposition 3.1 (On-diagonal and survival bounds)**.** *Suppose that $S \subset W$ is finite and that $\bigcup_{v\in S} P_v \Subset D$. There is a constant $C_d < \infty$, independent of S, such that, for every $x \in S$ and every $t > 0$,*

$$p_{2t}^S(x,x) \leq C_d\big(t^{-d/2} + \epsilon^d t^{-d}\big). \tag{30}$$

Consequently, whenever $t \geq \epsilon^2$,

$$\mathbb{P}_x(\tau_S > t) \leq C_d m(S)^{1/2} t^{-d/4}, \qquad m(S) := \sum_{v\in S} m_v. \tag{31}$$

Proof. Fix $x \in S$ and define

$$u_t(y) := p_t^S(x,y), \qquad X(t) := \|u_t\|_{2,m}^2.$$

By symmetry and the semigroup identity in (29),

$$X(t) = \sum_{y\in S} m_y p_t^S(x,y)^2 = p_{2t}^S(x,x). \tag{32}$$

Let $\mathcal{L}_S$ denote the killed generator, meaning that u_t is extended by zero outside S before (27) is applied. Self-adjointness and the weighted summation-by-parts identity give

$$X'(t) = 2\langle u_t, \mathcal{L}_S u_t\rangle_{L^2(m)} = -2\mathcal{D}(u_t). \tag{33}$$

For completeness, the last identity follows by pairing the two orientations of every edge:

$$-\sum_v m_v u(v)(\mathcal{L}_S u)(v) = \sum_{\{v,w\}\in\mathcal{E}} a(v,w)(u(v)-u(w))^2.$$

Also,

$$\|u_t\|_{1,m} = \sum_{y\in S} m_y p_t^S(x,y) = \mathbb{P}_x(t<\tau_S) \le 1. \tag{34}$$

Set

$$Q(t) := \mathcal{D}(u_t) = -\frac{1}{2}X'(t). \tag{35}$$

Applying (21) to u_t and using (34) gives

$$X(t)^{2+2/d} \le C_d Q(t)(X(t)+\epsilon^2 Q(t)). \tag{36}$$

We next extract a scalar differential inequality from (36). If $X(t)\ge \epsilon^2 Q(t)$, then

$$X(t)+\epsilon^2 Q(t) \le 2X(t),$$

so (36) gives

$$Q(t) \ge c_d X(t)^{1+2/d}. \tag{37}$$

If $X(t) < \epsilon^2 Q(t)$, then

$$X(t)+\epsilon^2 Q(t) < 2\epsilon^2 Q(t),$$

and (36) gives

$$Q(t) \ge c_d \epsilon^{-1} X(t)^{1+1/d}. \tag{38}$$

Using (35), the two cases imply

$$-X'(t) \ge c_d \min\left\{X(t)^{1+2/d},\, \epsilon^{-1}X(t)^{1+1/d}\right\}. \tag{39}$$

The terms inside the minimum agree exactly at $X=\epsilon^{-d}$.

Define the first threshold-crossing time

$$\sigma := \inf\{t\ge 0 : X(t)\le \epsilon^{-d}\}. \tag{40}$$

The function X is continuous and nonincreasing by (33). Thus $X(t)>\epsilon^{-d}$ for $0\le t<\sigma$, while $X(t)\le\epsilon^{-d}$ for $t\ge\sigma$. If $0<t<\sigma$, then the smaller term in (39) is $\epsilon^{-1}X(t)^{1+1/d}$, and hence

$$\frac{\mathrm{d}}{\mathrm{d}t}X(t)^{-1/d} = \frac{1}{d}X(t)^{-1-1/d}(-X'(t)) \ge c_d\epsilon^{-1}. \tag{41}$$

Integrating from 0 to $t<\sigma$ gives

$$X(t)^{-1/d} \ge X(0)^{-1/d} + c_d\epsilon^{-1}t \ge c_d\epsilon^{-1}t,$$

so

$$X(t) \le C_d \epsilon^d t^{-d}, \qquad 0<t<\sigma. \tag{42}$$

This estimate also shows that

$$\sigma \le C_d\epsilon^2. \tag{43}$$

Indeed, choose K_d so large that $C_d K_d^{-d}\le 1$. If $\sigma > K_d\epsilon^2$, then (42) at $t=K_d\epsilon^2$ would give $X(t)\le\epsilon^{-d}$, contradicting the definition of σ.

For $t\ge\sigma$, the smaller term in (39) is $X(t)^{1+2/d}$. Therefore

$$\frac{\mathrm{d}}{\mathrm{d}t}X(t)^{-2/d} = \frac{2}{d}X(t)^{-1-2/d}(-X'(t)) \ge c_d. \tag{44}$$

Integrating from σ to t and using $X(\sigma) \le \epsilon^{-d}$ gives

$$X(t) \le C_d\big(\epsilon^2 + t - \sigma\big)^{-d/2}, \qquad t \ge \sigma. \tag{45}$$

If $t \ge 2\sigma$, then $t - \sigma \ge t/2$, so (45) gives $X(t) \le C_d t^{-d/2}$. If $\sigma \le t < 2\sigma$, then monotonicity and (43) give

$$X(t) \le \epsilon^{-d} \le C_d \epsilon^d t^{-d}.$$

Together with (42), these estimates prove

$$X(t) \le C_d(t^{-d/2} + \epsilon^d t^{-d})$$

for every $t > 0$. The on-diagonal bound (30) now follows from (32).

Finally, Cauchy–Schwarz and (32) imply

$$\begin{aligned}\mathbb{P}_x(\tau_S > t) &= \sum_{y\in S} m_y p_t^S(x,y) \\ &\le m(S)^{1/2}\left(\sum_{y\in S} m_y p_t^S(x,y)^2\right)^{1/2} \\ &= m(S)^{1/2} p_{2t}^S(x,x)^{1/2}.\end{aligned}$$

When $t \ge \epsilon^2$,

$$\epsilon^d t^{-d} = t^{-d/2}\left(\frac{\epsilon^2}{t}\right)^{d/2} \le t^{-d/2}.$$

Applying (30) proves (31). □

4. The local tile-volume clock

The next construction supplies a Euclidean lower bound on the auxiliary clock. Although the tiling is not assumed to be given by a power diagram, the orthogonality relation implies that compatible power weights exist locally.

Lemma 4.1 (Local power potential)**.** *Let $B_R(z) \Subset D$. Define a piecewise-affine vector field on the ball by*

$$Z(y) := 2(y - v), \qquad y \in \operatorname{int} P_v \cap B_R(z). \tag{46}$$

Then there is a function $\psi \in W^{1,\infty}(B_R(z))$ such that $\nabla\psi = Z$ almost everywhere. For every cell whose interior meets $B_R(z)$, there is a constant $\omega_v \in \mathbb{R}$ such that

$$\psi(y) = \|y - v\|^2 - \omega_v, \qquad y \in P_v \cap B_R(z). \tag{47}$$

Moreover,

$$\|\nabla\psi\|_{L^\infty(B_R(z))} \le 2\epsilon. \tag{48}$$

Proof. We first prove that Z is curl-free in the sense of distributions. Inside each cell, its derivative is the symmetric matrix $2I$. Across a facet $H_e = P_v \cap P_w$,

$$Z|_{P_w} - Z|_{P_v} = 2(v - w),$$

which is parallel to the normal of H_e. Therefore the tangential component of Z has the same trace from the two sides. Cellwise integration by parts against a compactly supported smooth test field shows that all interface contributions to $\partial_i Z_j - \partial_j Z_i$ vanish. Hence

$$\partial_i Z_j - \partial_j Z_i = 0 \quad \text{in } \mathcal{D}'(B_R(z)) \quad \text{for all } i, j. \tag{49}$$

For completeness, we construct the potential from (49). Fix $0 < \rho < R$. Choose $\chi \in C_c^\infty(B_R(z))$ which equals one on a neighborhood of $\overline{B_\rho(z)}$, extend χZ by zero to $\mathbb{R}^d$, and let η_δ be a standard mollifier. If δ is smaller than the distance from $B_\rho(z)$ to the set where $\chi \neq 1$, then

$$Z_\delta := (\chi Z) * \eta_\delta$$

is smooth on $B_\rho(z)$. On this ball its distributional curl is the convolution of the distributional curl of Z with η_δ, so it is zero by (49). Since the ball is convex, the function

$$\psi_\delta(x) := \int_0^1 Z_\delta(z + s(x-z)) \cdot (x-z)\, \mathrm{d}s$$

satisfies $\nabla \psi_\delta = Z_\delta$ and $\psi_\delta(z) = 0$. Furthermore,

$$\|\nabla \psi_\delta\|_{L^\infty(B_\rho(z))} \leq \|Z\|_{L^\infty(B_R(z))}.$$

After passing to a subsequence, the Arzelà–Ascoli theorem gives uniform convergence on compact subballs to a Lipschitz function ψ, and $Z_\delta \to Z$ in L^1_{loc}. It follows distributionally that $\nabla \psi = Z$. A diagonal argument in ρ gives a single function on $B_R(z)$.

For almost every $y \in P_v \cap B_R(z)$,

$$\nabla\big(\psi(y) - \|y - v\|^2\big) = 0.$$

The set $P_v \cap B_R(z)$ is convex and hence connected, so this difference is constant there. Taking the continuous representative of the Lipschitz function ψ gives (47). Finally, if $y \in P_v$, then $\|y - v\| \leq \operatorname{diam}(P_v) \leq \epsilon$, so (46) gives (48). □

Lemma 4.2 (Exact clock identity)**.** *Suppose that $B_{r+3\epsilon}(z) \Subset D$, and choose the weights (ω_v) from Lemma 4.1 on this ball. Define*

$$\Phi_z(v) := \|v - z\|^2 - \omega_v. \tag{50}$$

Then, for every $v \in \mathcal{V} \cap B_r(z)$,

$$\mathcal{L}\Phi_z(v) = 2d. \tag{51}$$

Moreover, if $v, w \in \mathcal{V} \cap B_{r+2\epsilon}(z)$, then

$$|\omega_v - \omega_w| \leq 2\epsilon \|v - w\|. \tag{52}$$

Proof. Let $e = \{v, w\}$ and choose $y \in H_e$ in the ball on which (47) is valid. Continuity of ψ gives

$$\|y - v\|^2 - \omega_v = \|y - w\|^2 - \omega_w.$$

A direct expansion therefore yields

$$\Phi_z(w) - \Phi_z(v) = 2(w - v) \cdot (y - z). \tag{53}$$

Now suppose $v \in B_r(z)$. Every point of P_v belongs to $B_{r+\epsilon}(z)$, and every neighbor w of v belongs to $B_{r+2\epsilon}(z)$. Thus (53) is available on every facet of P_v. Let c_e be the centroid of H_e and let

$$n_{v,w} := \frac{w - v}{\|w - v\|}$$

be the unit normal pointing out of P_v across H_e. Since the normal component of $y - z$ is constant on H_e, (1) and (53) imply

$$\begin{aligned}\Delta_a^{\mathcal{G}}\Phi_z(v) &= 2\sum_{w\sim v} b(e)n_{v,w}\cdot(c_e - z)\\ &= 2\int_{\partial P_v}(y-z)\cdot n(y)\,\mathrm{d}\mathcal{H}^{d-1}(y)\\ &= 2d\,\mathrm{vol}_d(P_v) = 2dm_v.\end{aligned}$$

The penultimate equality is the divergence theorem. Dividing by m_v proves (51).

For the second assertion, (47) evaluated at the marked points gives $\psi(v) = -\omega_v$ and $\psi(w) = -\omega_w$. The segment from v to w lies in $B_{r+2\epsilon}(z)$ because this ball is convex. Hence (48) and the fundamental theorem for Lipschitz functions on line segments give

$$|\omega_v - \omega_w| = |\psi(v) - \psi(w)| \le 2\epsilon\|v - w\|.$$

□

Proposition 4.3 (A fixed ball cannot be crossed instantly)**.** *Let $\widetilde{X}$ be the finite-state process with generator* (27)*, started from a vertex z. Let*

$$\tau_{B_r} := \inf\{t \ge 0 : \widetilde{X}_t \notin B_r(z)\}.$$

Assume that $B_{r+3\epsilon}(z) \Subset D$, and choose the local weights (ω_v) and the function Φ_z as in Lemma 4.2. Let τ be any stopping time. Then

$$\mathbb{P}_z(\tau_{B_r} < t \wedge \tau) \le \frac{2dt + 2\epsilon(r+2\epsilon)}{r^2}. \tag{54}$$

Proof. Extend Φ_z arbitrarily to the rest of the finite state space. This does not change (51), since every neighbor of a vertex in $B_r(z)$ lies in $B_{r+2\epsilon}(z)$, where the original definition of Φ_z is available. Set $T := t \wedge \tau_{B_r} \wedge \tau$. Since $T \le t$, the finite-state Dynkin formula [5, Chapter 2], applied to the extended function and (51), gives

$$\mathbb{E}_z[\Phi_z(\widetilde{X}_T)] = \Phi_z(z) + 2d\mathbb{E}_z[T]. \tag{55}$$

Since $\Phi_z(z) = -\omega_z$, this identity is equivalent to

$$\mathbb{E}_z\|\widetilde{X}_T - z\|^2 = 2d\mathbb{E}_z[T] + \mathbb{E}_z[\omega_{\widetilde{X}_T} - \omega_z]. \tag{56}$$

Every jump has length at most 2ϵ. Therefore $\widetilde{X}_T \in B_{r+2\epsilon}(z)$, and (52) gives

$$|\omega_{\widetilde{X}_T} - \omega_z| \le 2\epsilon\|\widetilde{X}_T - z\| \le 2\epsilon(r+2\epsilon).$$

Using $T \le t$ in (56), we obtain

$$\mathbb{E}_z\|\widetilde{X}_T - z\|^2 \le 2dt + 2\epsilon(r+2\epsilon). \tag{57}$$

On the event $\{\tau_{B_r} < t \wedge \tau\}$, one has $T = \tau_{B_r}$ and $\|\widetilde{X}_T - z\| \ge r$. Hence

$$r^2\mathbb{P}_z(\tau_{B_r} < t \wedge \tau) \le \mathbb{E}_z\|\widetilde{X}_T - z\|^2.$$

Combining this with (57) proves (54). □

Remark 4.4. The normalization in (27) approximates Δ, rather than $\frac{1}{2}\Delta$. This is why the corrected squared displacement in (56) grows at rate $2d$. Replacing the rates by $a(v,w)/(2m_v)$ changes only the auxiliary time scale.

5. Energy and weighted L^2 error

The required estimate is a standard finite-volume consistency bound; see Eymard et al. [3, Theorem 9.3] for the general finite-volume framework. We use the orthogonal-tiling form proved by Bou-Rabee and Gwynne [2, Proposition 2.6], and no part of that energy estimate is reproved here. Fix a bounded open set K such that

$$\overline{U} \subset K \Subset D, \tag{58}$$

and suppose that h_C is C^2 and harmonic on a neighborhood of $\overline{K}$. Put

$$\delta_K := \operatorname{dist}(\overline{U}, K^c) > 0. \tag{59}$$

If $v \in \mathcal{V}[U]$, then $P_v \subset B_\epsilon(U)$. If $w \in \partial\mathcal{V}[U]$ is adjacent to $v \in \mathcal{V}[U]$, then $\|w - v\| \le 2\epsilon$ and hence $P_w \subset B_{3\epsilon}(U)$. Consequently,

$$3\epsilon < \delta_K \quad \Longrightarrow \quad \widehat{U} \subset K. \tag{60}$$

For an edge $e = \{v, w\}$, set

$$Q_e := \operatorname{Conv}(v, H_e) \cup \operatorname{Conv}(w, H_e). \tag{61}$$

By Bou-Rabee and Gwynne [2, Lemma 2.4],

$$\|e\| b(e) = d \operatorname{vol}_d(Q_e). \tag{62}$$

The inclusion $\bigcup_{e\in\overline{\mathcal{E}[U]}} Q_e \subset \widehat{U}$ is Bou-Rabee and Gwynne [2, Lemma 2.5]. We need the corresponding bound for the sum of volumes. For a fixed v, the interiors of the pyramids $\operatorname{Conv}(v, H_e)$, $e \ni v$, are pairwise disjoint and their union is P_v up to a null set. Pyramids belonging to distinct cells also have disjoint interiors. Therefore all oriented pyramids which occur in the sets Q_e, $e \in \overline{\mathcal{E}[U]}$, have pairwise disjoint interiors, and

$$\sum_{e\in\overline{\mathcal{E}[U]}} \operatorname{vol}_d(Q_e) \le \sum_{v\in\overline{\mathcal{V}[U]}} \operatorname{vol}_d(P_v) = \operatorname{vol}_d(\widehat{U}) \le \operatorname{vol}_d(K), \tag{63}$$

where the last inequality uses (60).

Lemma 5.1 (Energy and weighted L^2 error)**.** *Let h_D solve* (7)*, let*

$$f := h_D - h_C \quad \text{on } \overline{\mathcal{V}[U]},$$

and extend f by zero to all of $\mathcal{V}$. Set

$$M := \|\nabla^2 h_C\|_{L^\infty(K)}.$$

Then, for all sufficiently small ϵ,

$$\mathcal{D}(f) + \|f\|_{2,m}^2 \le C_{d,K} M^2 \epsilon^2. \tag{64}$$

Proof. The boundary values in (7) imply that $f = 0$ on $\partial\mathcal{V}[U]$, so its global energy equals $\mathcal{D}(f, U)$. The finite-volume consistency estimate [2, Proposition 2.6] gives

$$\mathcal{D}(f) \le 9M^2\epsilon^2 \sum_{e\in\overline{\mathcal{E}[U]}} d \operatorname{vol}_d(Q_e). \tag{65}$$

Combining (65) with (63) gives

$$\mathcal{D}(f) \le 9d \operatorname{vol}_d(K) M^2 \epsilon^2. \tag{66}$$

The support of f is contained in $\mathcal{V}[U]$, so

$$V_f = \sum_{v:f(v)\neq 0} m_v \le \mathrm{vol}_d(\widehat{U}) \le \mathrm{vol}_d(K).$$

Applying the support inequality (23) and then (66), and using $\epsilon \le 1$, gives

$$\|f\|_{2,m}^2 \le C_{d,K}\mathcal{D}(f) \le C_{d,K}M^2\epsilon^2.$$

This proves (64). □

6. Uniform convergence from expected exit times

We first combine the support inequality with heat-kernel smoothing to bound the expected exit time of a set in terms of its tile volume. This estimate will be applied to a superlevel set of the Dirichlet error.

Lemma 6.1 (Expected exit time from a finite set)**.** *Let $S \subset \mathcal{V}$ be a nonempty finite set with $\bigcup_{v\in S} P_v \Subset D$, and let τ_S be the exit time of the finite-state process with generator (27). Set*

$$R = m(S)^{1/d} = \left(\sum_{v\in S} m_v\right)^{1/d}, \qquad \rho = R^2 + \epsilon R. \tag{67}$$

Then

$$\sup_{x\in S} \mathbb{E}_x \tau_S \le C_d(R^2 + \epsilon R). \tag{68}$$

Proof. For functions supported on S, inequality (23) gives $\|f\|_{2,m}^2 \le C_d\rho\mathcal{D}(f)$. The smallest eigenvalue of the negative killed generator therefore satisfies

$$\lambda_1(S) \ge c_d/\rho. \tag{69}$$

In particular the killed generator is strictly positive, so the exit time is almost surely finite. Let P_t^S be its killed semigroup and $p_t^S(x,y)$ the density with respect to m. Proposition 3.1, equation (30), gives

$$\begin{aligned} m(S)p_{2\rho}^S(x,x) &\le C_d R^d\big(\rho^{-d/2} + \epsilon^d\rho^{-d}\big) \\ &= C_d\left[\left(\frac{R^2}{\rho}\right)^{d/2} + \left(\frac{\epsilon R}{\rho}\right)^d\right] \le C_d. \end{aligned} \tag{70}$$

For $t \ge \rho$, reversibility, the semigroup property, and $L^2(m)$ spectral contraction imply

$$\begin{aligned} \mathbb{P}_x(\tau_S > t) &= \langle p_\rho^S(x,\cdot), P_{t-\rho}^S \mathbf{1}_S\rangle_m \\ &\le \big[p_{2\rho}^S(x,x)m(S)\big]^{1/2} e^{-\lambda_1(S)(t-\rho)} \\ &\le C_d \exp\left(-\frac{c_d(t-\rho)}{\rho}\right). \end{aligned} \tag{71}$$

Integrating the survival probability over $[0,\rho]$ and $[\rho,\infty)$ proves (68). □

Proof of Theorem 1.1. Write $h = h_C$ and $f = h_D - h_C$. The Dirichlet problem is linear, so it suffices first to prove the estimate under the normalization $\|h\|_{C^2(K)} \le 1$. If that norm is zero, the maximum principle already gives $f = 0$; otherwise rescale h_C and h_D by the same positive constant. Fix $r > 0$, depending only on U and K, so that $B_{4r}(\overline{U}) \Subset K$, and take ϵ sufficiently small. In particular all stopped positions below lie in K, and the maximum principle gives $|h_D| \le 1$ on $\overline{\mathcal{V}[U]}$.

Fix $0 < \eta \le 1$ and set

$$S = \{v \in \mathcal{V}[U] : f(v) > \eta/2\}. \tag{72}$$

If S is empty then the positive error is at most $\eta/2$. Otherwise Lemma 5.1, equation (64), and Chebyshev's inequality imply

$$m(S) \le C_{d,K}\epsilon^2\eta^{-2}, \qquad R = m(S)^{1/d} \le C_{d,K}(\epsilon/\eta)^{2/d}. \tag{73}$$

The cells of S have compact union in D for small mesh, so Lemma 6.1 applies.

Let $z \in \mathcal{V}[U]$ maximize f. We may assume $f(z) > \eta/2$, so $z \in S$. Run the process (27) on the finite set $\overline{\mathcal{V}[U]}$, with $\partial\mathcal{V}[U]$ absorbing. Let τ_U be its absorption time, let τ_{B_r} be the first exit from $B_r(z)$, and set

$$\sigma = \tau_S \wedge \tau_{B_r}, \qquad E = \{\tau_{B_r} < \tau_S\}. \tag{74}$$

Since $f = 0$ on $\partial\mathcal{V}[U]$, we have $\sigma \le \tau_S \le \tau_U$; furthermore $\mathbb{E}_z\sigma \le C_d(R^2 + \epsilon R)$.

The coordinate functions are discrete harmonic. Indeed, for each interior vertex v, orthogonality and the divergence theorem give

$$\sum_{w\sim v} a(v,w)(w - v) = \sum_{e\ni v} b(e)n_{v,w} = \int_{\partial P_v} n \,\mathrm{d}\mathcal{H}^{d-1} = 0.$$

The stopped coordinates remain in $B_{r+2\epsilon}(z)$ because every jump has length at most 2ϵ. Optional stopping, first at $\sigma \wedge N$ and then by bounded convergence, therefore yields

$$\mathbb{E}_z[\widetilde{X}_\sigma - z] = 0. \tag{75}$$

Use the local weights of Lemma 4.2 on $B_{r+3\epsilon}(z) \Subset K$. The same finite-state Dynkin identity as (55)–(56), first at $\sigma \wedge N$, gives in the limit

$$\begin{aligned}\mathbb{E}_z\left\|\widetilde{X}_\sigma - z\right\|^2 &= 2d\mathbb{E}_z\sigma + E_z[\omega_{\widetilde{X}_\sigma} - \omega_z] \\ &\le 2d\mathbb{E}_z\sigma + 2\epsilon(r + 2\epsilon) \\ &\le C_{d,U,K}(R^2 + \epsilon R + \epsilon).\end{aligned} \tag{76}$$

Here the passage to the limit uses the finite mean of σ, the bounded stopped positions, and the local weight bound (52).

Taylor's theorem on the ball containing the stopped positions and (75) imply

$$\mathbb{E}_z[h(\widetilde{X}_\sigma) - h(z)] \le \tfrac{1}{2}\left\|\nabla^2 h\right\|_{L^\infty(K)} \mathbb{E}_z\left\|\widetilde{X}_\sigma - z\right\|^2. \tag{77}$$

Also, on E the stopped displacement has size at least r, so

$$\mathbb{P}_z(E) \le r^{-2}\mathbb{E}_z\left\|\widetilde{X}_\sigma - z\right\|^2. \tag{78}$$

Since $h_D(\widetilde{X}_{t\wedge\tau_U})$ is a bounded martingale, optional stopping gives

$$h_D(z) = \mathbb{E}_z h_D(\widetilde{X}_\sigma).$$

On E^c, the terminal vertex is outside S and its error is at most $\eta/2$; on E, the maximum principle and $|h| \le 1$ bound its error by 2. Therefore

$$\begin{aligned} f(z) &= \mathbb{E}_z f(\widetilde{X}_\sigma) + \mathbb{E}_z[h(\widetilde{X}_\sigma) - h(z)] \\ &\le \eta/2 + 2\mathbb{P}_z(E) + C\mathbb{E}_z \left\| \widetilde{X}_\sigma - z \right\|^2 \\ &\le C_{d,U,K}(\eta + R^2 + \epsilon R + \epsilon). \end{aligned} \tag{79}$$

This also covers the case $f(z) \le \eta/2$. Apply the same argument to $-h_C$ and its discrete extension $-h_D$. Inserting (73) yields

$$\|h_D - h_C\|_\infty \le C_{d,U,K} \left[\eta + (\epsilon/\eta)^{4/d} + \epsilon(\epsilon/\eta)^{2/d} + \epsilon \right]. \tag{80}$$

Choose $\eta = \epsilon^{4/(d+4)}$. The first two terms are then equal; the other two are at most a constant times the same power for $0 < \epsilon \le 1$ and $d \ge 2$. Restoring the original normalization gives

$$\max_{v \in \mathcal{V}[U]} |h_D(v) - h_C(v)| \le C_{d,U,K} \, \|h_C\|_{C^2(K)} \, \epsilon^{4/(d+4)}. \tag{81}$$

This proves (10) with $\beta_d = 4/(d+4)$. For the qualitative conclusion (9), choose a bounded open K containing $\overline{U}$, with closure inside the neighborhood on which h_C is harmonic. This completes the proof. □

7. Deduction of the random-walk consequences

Proof of Corollary 1.2. Let D_0 denote the domain called D in Assumption D of Bou-Rabee and Gwynne [2, Section 4], and let D_{amb} be the ambient domain tiled by the graphs. Choose $\delta > 0$ such that

$$B_{2\delta}(\overline{D_0}) \Subset D_{\mathrm{amb}}.$$

For $r > 0$ and $x \in B_\delta(D_0)$, set

$$B := B_r(x) \cap B_\delta(D_0),$$

exactly as in that assumption. This is a bounded domain with $B \Subset D_{\mathrm{amb}}$.

We first verify Bou-Rabee and Gwynne [2, Assumption D, equation (4.1)]. If $y \in B$ and $y \in P_v$, then

$$\operatorname{dist}(y, V_n) \le \|y - v\| \le \operatorname{diam}(P_v) \le \epsilon_n.$$

Hence the nearest-vertex distance tends to zero uniformly on B. Next let h_C be harmonic on a neighborhood of $\overline{B}$. Applying Theorem 1.1 with $U = B$ gives Bou-Rabee and Gwynne [2, Assumption D, equation (4.2)]. Thus Assumption D is verified.

Under Assumption D, Bou-Rabee and Gwynne [2, Proposition 4.2] gives uniform convergence of harmonic measure on balls. The proof of Bou-Rabee and Gwynne [2, Theorem 4.1] then concatenates successive ball exits by the strong Markov property

and yields convergence of the stopped canonical walk modulo time parametrization. Finally, the argument labeled "Proof of Theorem B assuming Hypothesis (b)" in Bou-Rabee and Gwynne [2, Section 4] transfers this random-walk convergence to the Dirichlet problem with continuous boundary data on a Lipschitz domain. These invocations use condition (1.3) and Assumption D, but not alternatives (I)–(III). □

Remark 7.1 (Why arbitrarily small cells cause no obstruction). The weighted L^2 estimate (64) forces a superlevel set of the Dirichlet error to have small tile volume. Lemma 6.1 then bounds the expected time spent in this set. The exact local clock identity and the coordinate martingale convert that time bound into the stopped-displacement estimate (76). Taylor's theorem and optional stopping give (80), whose optimization yields (81). In dimension two the resulting harmonic-extension rate is $\epsilon^{2/3}$. The walk conclusion above remains a convergence statement modulo time parametrization under the remaining hypotheses of Bou-Rabee and Gwynne; no rate for a parametrized walk is asserted.

References

[1] L. Ambrosio, N. Fusco, and D. Pallara. *Functions of Bounded Variation and Free Discontinuity Problems*. Oxford Mathematical Monographs. Clarendon Press, Oxford, 2000.

[2] A. Bou-Rabee and E. Gwynne. *Random walk on sphere packings and Delaunay triangulations in arbitrary dimension*. Proceedings of the London Mathematical Society, 131:e70084, 2025. doi:10.1112/plms.70084; arXiv:2405.11673v2.

[3] R. Eymard, T. Gallouët, and R. Herbin. Finite volume methods. In P. G. Ciarlet and J.-L. Lions, editors, *Handbook of Numerical Analysis*, volume VII, pages 713–1020. North-Holland, Amsterdam, 2000.

[4] G. B. Folland. *Real Analysis: Modern Techniques and Their Applications*. Wiley, New York, second edition, 1999.

[5] J. R. Norris. *Markov Chains*. Cambridge Series in Statistical and Probabilistic Mathematics, vol. 2. Cambridge University Press, Cambridge, 1998.